La Scuola d'Italia / Dr. Huson / IB Math 12
26 October 2026

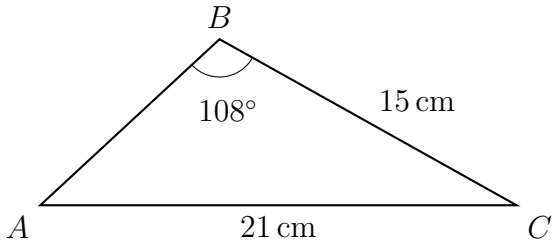
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2.4 Pre-Quiz: Sine Rule, Cosine Rule and the Area of a Triangle

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad c^2 = a^2 + b^2 - 2ab\cos C \qquad \text{Area} = \frac{1}{2}ab\sin C$$

A calculator is allowed. Give answers correct to three significant figures, unless they are exact. When a question gives an angle in radians, give angles in radians. Diagrams are not to scale.

1. The following diagram shows triangle ABC, with $AC = 21\,\text{cm}$, $BC = 15\,\text{cm}$ and $A\hat{B}C = 108°$.

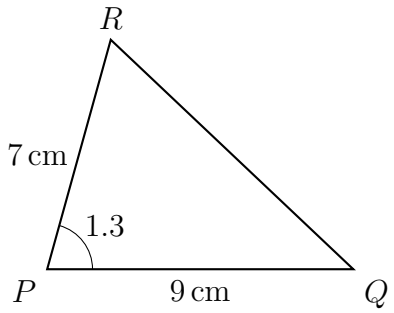


(a) Find $B\hat{A}C$.

(b) Find AB.

(c) Find the area of triangle ABC.

2. The following diagram shows triangle PQR, with $Q\hat{P}R = 1.3$ radians, $PQ = 9\,\text{cm}$ and $PR = 7\,\text{cm}$.



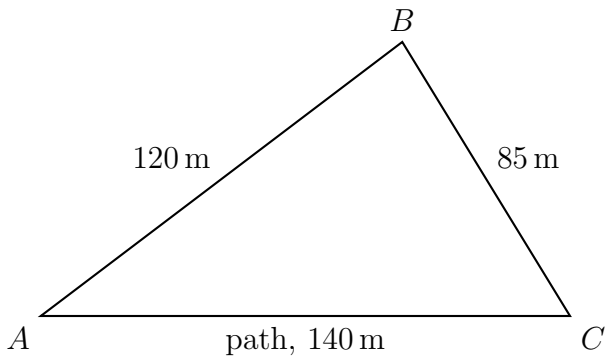
(a) Find QR.

(b) Find the area of triangle PQR.

(c) Find $P\hat{Q}R$.

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3. A farmer owns a triangular field ABC, with $AB = 120\,\text{m}$, $BC = 85\,\text{m}$ and $AC = 140\,\text{m}$. A straight path runs along the side AC.



(a) Find $A\hat{B}C$.

(b) Find the area of the field.

(c) The farmer walks from B to the path by the shortest possible route. Find the length of this route.

4. In triangle ABC, $AB = 10\,\text{cm}$, $BC = 7\,\text{cm}$ and $B\hat{A}C = 0.6$ radians.

   (a) Find the two possible values of $A\hat{C}B$.

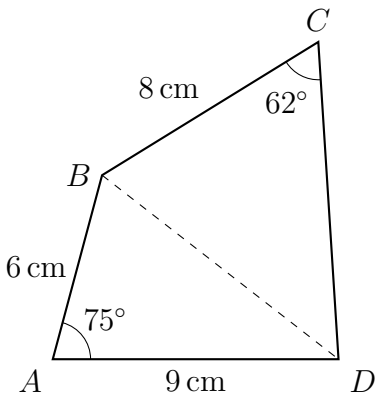
   In the rest of this question, $A\hat{C}B$ is obtuse.

   (b) Find $A\hat{B}C$.

   (c) Find the area of triangle ABC.

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5. The following diagram shows quadrilateral $ABCD$, with $AB = 6\,\text{cm}$, $AD = 9\,\text{cm}$, $BC = 8\,\text{cm}$, $B\hat{A}D = 75°$ and $B\hat{C}D = 62°$.

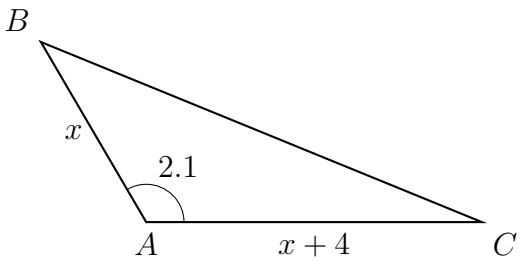


(a) Find BD.

(b) Find $B\hat{D}C$.

(c) Find the area of quadrilateral $ABCD$.

6. The following diagram shows triangle ABC, with $AB = x$ cm, $AC = (x + 4)$ cm and $B\hat{A}C = 2.1$ radians. The area of triangle ABC is $30\,\text{cm}^2$.



(a) Find the value of x.

(b) Find BC.

(c) Find $A\hat{B}C$.